

ASHUTOSH RAVAL

Senior Full-Stack Backend Developer

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PROFESSIONAL SUMMARY

Results-driven Senior Node.js Developer with **6+ years of professional experience** designing and implementing scalable backend architectures for diverse, high-traffic applications. Proven expertise in building microservices, event-driven systems, and real-time applications serving 100k+ concurrent users. Strong background in cloud infrastructure (AWS, Google Cloud, Firebase), message streaming (Apache Kafka), and payment gateway integration. Passionate about writing clean, maintainable code and mentoring engineering teams to deliver production-grade solutions.

CORE COMPETENCIES

Backend & Framework Technologies

- **Node.js Frameworks & Patterns:** Event-driven architecture, microservices, REST APIs, GraphQL
- **Languages & Ecosystems:** JavaScript (ES6+), TypeScript, Node.js runtime optimization
- **Real-time Communication:** [Socket.IO](#), WebSockets, Redis Pub/Sub for live notifications and presence tracking

Application Development Domains

- Order Management Systems (multi-vendor, inventory sync, fulfillment workflows)
- Real-time Chat & Messaging Applications (horizontal scaling, concurrent user handling)
- Transportation & Logistics (ride tracking, SIM-based GPS integration, shipment management)
- Content Management Systems (CMS architecture, dynamic publishing)
- Insurance & Policy Management (policy underwriting, claims processing, document handling)
- Dating Applications (matching algorithms, real-time interactions, content moderation)
- Slot Booking & Multi-tenant Platforms (concurrency control, dynamic scheduling)

Cloud & Infrastructure

- **AWS:** Lambda (serverless), S3, EC2, CloudWatch, IAM, SSO/OAuth integrations
- **Google Cloud:** Cloud Functions, Firestore, Cloud Storage, Pub/Sub messaging
- **Firebase:** Authentication, Realtime Database, Cloud Firestore, Hosting

- **DevOps Tools:** Docker, Docker Compose, containerization, NPM/Yarn package management

Data & Messaging

- **Message Streaming:** Apache Kafka, KafkaJS for event processing and distributed systems
- **Databases:** MongoDB (document modeling, aggregation pipelines), SQL (relational design), PostgreSQL
- **Caching & Performance:** Redis (caching strategies, distributed sessions, pub/sub, rate limiting)
- **NoSQL Patterns:** Firestore, document-based architectures for unstructured data

Enterprise Integrations

- **Payment Gateways:** Stripe, Razorpay, Paytm (payment processing, subscription billing, payouts)
- **Authentication:** Passport.js, JWT, session management, multi-factor authentication
- **Third-Party APIs:** RESTful integrations, webhook handling, API rate limiting
- **CMS Platforms:** Strapi, Payload CMS for headless content delivery
- **ORMs & Query Builders:** Prisma, TypeGraphQL for type-safe database operations
- **Tracking & Monitoring:** Traqo (GPS tracking), production monitoring, error tracking

PROFESSIONAL EXPERIENCE

Senior Node.js Developer

Webelight Solution Pvt. Ltd., Ahmedabad, India | July 2018 – Present (7+ Years)

Leadership & Architecture

- Led architectural design and code reviews for multiple microservice-based projects, ensuring scalability and adherence to best practices
- Managed sprint planning, task allocation, and technical mentorship for junior developers
- Defined performance requirements and SLAs for backend systems serving high-traffic production environments
- Provided accurate time estimates and delivered projects on schedule

Backend Development & Microservices

- Designed and implemented microservices architecture enabling independent deployment and scaling of application components
- Built event-driven systems processing real-time data streams with Kafka and [Socket.IO](#)
- Developed robust REST and GraphQL APIs using modern Node.js patterns (middleware, dependency injection, error handling)
- Optimized database queries and caching strategies to reduce latency and improve application performance

Real-Time & Scalability

- Engineered horizontally-scalable chat microservice supporting 100k+ concurrent users with load balancing
- Implemented Redis pub/sub for real-time presence updates, notifications, and inter-service communication
- Optimized message pipelines to handle high-throughput scenarios with minimal latency

Cloud & DevOps

- Deployed applications on AWS (Lambda for serverless workloads, EC2 for compute, S3 for storage)
- Integrated Google Cloud and Firebase services for authentication, real-time databases, and hosted backends
- Implemented Docker containerization for consistent development and production environments
- Set up CI/CD pipelines and monitoring for production-grade reliability

Quality Assurance

- Developed comprehensive test suites achieving 95% code coverage across critical paths
- Implemented health-check endpoints, logging, and alerting for production monitoring
- Maintained code security and vulnerability management in payment-critical systems

KEY PROJECT HIGHLIGHTS

Chat Microservice - Real-Time Messaging Platform

- **Technology Stack:** Node.js, MongoDB, Redis, GraphQL, [Socket.IO](#), Docker, AWS
- **Scale:** 100k+ concurrent users with sub-100ms latency for message delivery
- **Key Achievements:**
 - Engineered horizontal scaling with load balancing across multiple instances
 - Implemented Redis Pub/Sub for distributed real-time presence and notification broadcasting
 - Optimized message pipeline to maintain performance under high concurrent load
 - Developed automated health checks and achieved 95% test code coverage
 - Built GraphQL subscription APIs for real-time data synchronization

Multi-Tenant Slot Booking Platform

- **Technology Stack:** Node.js, MongoDB, Razorpay, Express.js, WebSockets
- **Business Model:** B2B SaaS with strict data isolation and tenant segmentation
- **Key Achievements:**
 - Designed multi-tenant architecture with row-level security and data isolation per client
 - Implemented optimistic concurrency control to prevent double-bookings in high-contention scenarios
 - Built dynamic slot allocation engine with real-time calendar synchronization

- Integrated Razorpay for payment processing and automated payout workflows
- Developed automated email/SMS confirmation system using third-party providers

Transportation & Freight Booking Platform (Navodaya Transport)

- **Technology Stack:** Node.js, MongoDB, Razorpay, Traqo GPS, Express.js, Admin Dashboard
- **Features:** Role-based access (Loader, Trucker, Admin), real-time shipment tracking
- **Key Achievements:**
 - Built role-based freight booking system with shipment request lifecycle management
 - Integrated SIM-based GPS tracking via Traqo for real-time location monitoring
 - Implemented end-to-end payment workflows for seamless booking-to-settlement process
 - Developed admin dashboard for user management, booking oversight, and payment reconciliation
 - Designed notification system for real-time booking and shipment status updates

Additional Projects

- **Order Management Systems:** Multi-vendor inventory sync, fulfillment workflows, payment reconciliation
- **Content-Based Applications:** CMS architecture with dynamic publishing, search, and recommendation engines
- **Insurance & Policy Management:** Policy underwriting, claims processing, document management
- **Dating Application:** User matching algorithms, real-time messaging, content moderation

TECHNICAL EXPERTISE SUMMARY

Category	Technologies
Backend Frameworks	Node.js (Express, Fastify, NestJS), Microservices, Event-Driven Architecture
Languages	JavaScript (ES6+), TypeScript, SQL
Databases	MongoDB, PostgreSQL, MySQL, Firebase Firestore, Realtime Database
Caching & Messaging	Redis, Apache Kafka, KafkaJS, Socket.IO , Pub/Sub, Message Queues
APIs	REST, GraphQL (Apollo, TypeGraphQL, Prisma), WebSockets
Cloud Platforms	AWS (Lambda, EC2, S3, CloudWatch, IAM), Google Cloud (Functions, Firestore, Pub/Sub), Firebase
Authentication	OAuth 2.0, Passport.js, JWT, Session Management, Multi-factor Authentication
Payment Integration	Stripe, Razorpay, Paytm (payment processing, subscriptions, payouts)
CMS & Headless	Strapi, Payload CMS
DevOps & Deployment	Docker, Docker Compose, NPM/Yarn, Git, CI/CD fundamentals
Monitoring & Tracking	Application logging, error tracking, GPS tracking (Traqo), health checks

PUBLICATIONS & THOUGHT LEADERSHIP

- **"In-App Server Notification for Android"** - Medium publication on real-time notification implementation
 - **"Running Danger.js Locally"** - Medium article on development workflow optimization
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EDUCATION

Master of Computer Applications (MCA)

Acharya Motibhai Patel Institute — Ganpat University, Gujarat | 2019

- SGPA: 8.3/10

Bachelor of Computer Applications (BCA)

Hemchandracharya North Gujarat University, Gujarat | 2015

- SGPA: 6.8/10

ADDITIONAL INFORMATION

- **Languages:** English, Hindi, Gujarati (Professional fluency)
- **Open Source Contributions:** Active in Node.js ecosystem
- **Continuous Learning:** Regular engagement with emerging backend technologies and architectural patterns
- **Problem Solving:** Known for debugging complex production issues and implementing optimized solutions